

Supplement — Accountable Neural Aggregation in Computational Pathology

Internal-review supplement (2026-08-04). Not for public release.

1 Implementation-path inventory

Table 1 lists the primary implementation paths per research line, bound in the evidence manifest (`evidence/manuscript_evidence_manifest.json`). Every architectural claim in the main text binds to these source paths and their tests.

Table 1: Implementation-path inventory.

Line	Primary implementation paths
PCam	<code>experiments/train_pcam.py</code> , <code>evaluate_pcam.py</code> , <code>src/data/datasets/pcam_dataset.py</code>
PA-NF	<code>experiments/paired_acquisition/runners.py</code> , <code>src/paired_acquisition_factorial.py</code>
SCORPION	<code>experiments/paired_acquisition/run.*scorpion*</code> , <code>src/models/scorpion_pathoalign.py</code>
Canine	<code>run_biological_label_preservation_fixed_estimand.py</code>
TransnnMIL	<code>src/models/transnnmil/</code> ; <code>src/models/mil/</code>
PathologyFL	<code>src/features/federated/pathology_fl/</code>
FAIR-WEIGHTS-H	<code>.../weighting/fair_weights.h.py</code> , <code>.../pathoalign_bridge.py</code>
CAMELYON17	<code>experiments/camelyon17_*</code> ; <code>scripts/camelyon17/</code>
Audit	<code>scripts/provenance/</code> ; <code>src/paired_acquisition_provenance.py</code>

2 Test coverage

Table 2 maps each line to its test suites. Tests are the binding target for architectural claims.

Table 2: Test coverage per research line.

Line	Test suites
PCam	<code>tests/test_pcam.*.py</code> , <code>test_pcam_evaluation.ci.py</code>
Paired/SCORPION/canine	<code>tests/test_fixed_estimand_real_feature_space_adjudication*.py</code> , <code>tests/test_real_paired_scanner_result.py</code>
Synthetic	<code>tests/test_biological_bottleneck_capacity_allocation_factorial.py</code>
TransnnMIL	<code>tests/models/test_transnnmil.*.py</code> , <code>test_topology_branch.py</code> , <code>test_hierarchical_pooling.py</code>
PathologyFL	<code>tests/federated/test_fl_integration.py</code> , <code>test_pathology_fl.*.py</code> , <code>test_secure_aggregation*.py</code>
FAIR-WEIGHTS-H	<code>tests/federated/test_fair_weights.h.py</code> , <code>test_weighting.*.py</code>
CAMELYON17	<code>tests/test_camelyon.*.py</code> , <code>tests/camelyon17/</code>
Audit	<code>tests/test_corrected_paired_acquisition_evidence.py</code> , <code>test_real_bottleneck_representation.py</code>

3 Result artifact hashes

The authoritative artifact hashes are recorded in `evidence/manuscript_evidence_manifest.json`. Selected current hashes:

Table 3: Selected result-artifact hashes (2026-08-04).

Artifact	SHA-256 (first 16)
Corrected evidence manifest	<code>a7a7e34d1df24f70</code>
Corrected snapshot	<code>cb06886a6050a66f</code>
Real paired-scanner result	<code>7293ddcddf8fbaf</code>
Recovery result	<code>2075a2aa240877a1</code>
Adjudication v2 result	<code>2cccaad7aab89586</code>

4 Per-fold PA-NF corrected metrics

Per-fold corrected five-category balanced accuracy, scanner balanced accuracy, and worst-pair retrieval for the recovered B32/B64 cells and the deterministic baselines are recorded in the adjudication v2 result (`canine_metric_table`). The v2 summary reports B64–B32 category mean -0.008 (one positive fold), scanner mean $+0.049$ (five of five), and worst-pair retrieval mean $+0.050$ (five of five).

5 Synthetic status inventory

Synthetic lines are classified `synthetic_mechanism_evidence`. The biological-bottleneck capacity-allocation factorial status is `complete_capacity_gain_with_scanner_tradeoff`. The corrected adjudication reports `synthetic_accessibility_evidence` = `false` because the required corrected category gain is absent.

6 TransnnMIL component matrix

Table 4: TransnnMIL component status matrix.

Component	Implemented	Tested	Empirical result
Canonical dual-branch	yes	yes	preliminary PANDA runs
Branch-token self-attention fusion	yes	yes	incremental contribution not measured
TransnnMILv2 (3-branch)	yes	yes	none
Concat/gate/learned-attention controls	yes	yes	planned controls, none
Hierarchical pooling	yes	yes	synthetic-MIL ablation only
Topology/GNN branch	yes	yes	historical withdrawn; no repaired rerun
Adaptive pruning	yes	yes	disabled in v2; none
Graph caching	yes	yes	none

7 PathologyFL feature matrix

PathologyFL implements FedAvg, FedProx, FedAdam, weighted, Byzantine-robust, secure (TenSEAL), pathology-aware, and FAIR-WEIGHTS-H aggregators; DP-SGD and privacy budget tracking; TLS gRPC and secure aggregation; monitoring and Byzantine detection; async training; fault tolerance; compression; and a model registry. The e2e federated tests have dangling imports and are not runnable as written; secure/DP guarantees are gated on optional libraries.

8 FAIR-WEIGHTS-H signal and constraint definitions

Implemented: integrity gate, uncertainty penalty, useful uniqueness, bounded-weight caps and renormalization, conservative mode, PathoAlign audit bridge (representation risk, reason codes, mechanism validity). Specification- only or partial: training/validation/monitoring weight separation, difficulty adjustment, Owen/Shapley contribution estimator, subgroup-performance constraints, weight-change limits, four-mode fallback table, validated drift detection.

9 CAMELYON17 boundary

CAMELYON17 weighting studies are centralized frozen-feature proxies for source-center influence on one held-out center. They are not full federated- learning validation, and one held-out center does not establish a cross-hospital law. Adversarial center removal was negative; a supervised linear center subspace was partially removable.

10 PANDA boundary

PANDA studies are simulated-site, cached-feature experiments over synthetic institutional partitions. There is no patient-level endpoint and no real hospital mapping. TransnnMIL historical QWK values are withdrawn; post-repair runs are preliminary and pending matched controlled reruns.

11 Historical withdrawal ledger

The authoritative ledger is CLAIM_BOUNDARY.md and docs/research/scientific-audit-remediation-20260725.md. Withdrawn families include: historical TransnnMIL fusion/topology results, historical canine biological-category audit metrics, SCORPION slide-level sign-flip p-values, unified-scoreboard rankings, PCam clinical/superiority claims, historical canine category metrics, and cosine-as-biology. Withdrawn artifacts never appear in active result tables.